

Automation Walkthrough

1. Introduction

This document explains the internal logic of the `Test_001_Registration` automation script, mapping user actions to code execution.

2. Automation Flow Checklist

Step	Portal Page	Automation Logic	Verification
1	Login/Basic Details	<code>RegistrationPage.set_name,</code> <code>set_dob</code>	Modal handled (<code>handle_modal</code>), UDISE verified.
2	Eligibility	<code>EligibilityPage.set_date_of_appointment</code> <code>contains</code>	Checks URL eligibility.
3	Authentication	<code>AuthenticationPage.set_email</code>	Uses incremental email logic (<code>DataUtils</code>) to ensure uniqueness.
4	OTP Verification	OTPPage (Manual Entry)	Script waits (<code>explicit_wait</code>) for user input.
5	Personal Info	<code>PersonalInformationPage</code>	Dropdowns selected using robust <code>BasePage</code> methods.
6	Address Details	<code>AddressDetailsPage.select</code>	dynamic wait for dropdown to populate.

7	Subject Details	SubjectDetailsPage	Auto-detects enabled subjects and selects "Any" -> "English/Hindi".
8	Documents	DocumentsPage.upload_all_documents DataUtils.	Uses dummy files generated by DataUtils.
9	Payment Review	PaymentFlowPage.check_all_confirmation_boxes	JS-based clicks to toggle checkboxes.
10	Gateway	PaymentFlowPage.enter_card_details	Switch to Frames, enter dummy card, simulate success.

3. Key Automations Explained

A. Dynamic Data Generation

Instead of hardcoding data, the framework uses `utilities/data_utils.py`:

- **Emails:** `insphere.shubhamsingh+N@gmail.com` (Where N increments every run).
- **Files:** `dummy.jpg` and `dummy.pdf` are created on the fly if missing.

B. Robust Dropdown Handling

The `BasePage.select_chosen_option` method handles the complex "Chosen.js" dropdowns used in the portal by:

1. Trying to click the "Chosen" UI container.
2. If that fails, unhiding the standard `<select>` element and interacting with it directly.
3. Supporting partial text matches (e.g., "Delhi" matches "GOVT OF DELHI").

C. Payment Gateway Interaction

The SabPaisa gateway often uses iframes. The `PaymentFlowPage` uses a smart retry loop:

1. Scans the page for frames.
2. Switches to each frame to look for card input fields (`cardNumber`).
3. Once found, enters data and switches back.

4. Execution Recording

To record the execution:

- The automation tool automatically attempts to record the session.
- Developers can use tools like OBS Studio to capture the screen during `python -m pytest` execution.